

Market Basket Analysis with Python - Snapdeal

Project Video Link:- <https://youtu.be/FHVIFFgDS1Q>

Project Link:- <https://colab.research.google.com/drive/1t95HK5bsx4-s1vD0mIXuQZ0GktoP2zQT?usp=sharing>

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn as sk

SnapDeal=pd.read_csv(r"D:\Internshala Projects\Project 5\
SnapDeal.csv")

SnapDeal.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 24 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Timestamp                                800 non-null    object
1   age                                       800 non-null    int64
2   Gender                                   800 non-null    object
3   Purchase_Frequency                      800 non-null    object
4   Purchase_Categories                     800 non-null    object
5   Personalized_Recommendation_Frequency  800 non-null    object
6   Browsing_Frequency                     800 non-null    object
7   Product_Search_Method                   651 non-null    object
8   Search_Result_Exploration               800 non-null    object
9   Customer_Reviews_Importance             800 non-null    int64
10  Add_to_Cart_Browsing                    800 non-null    object
11  Cart_Completion_Frequency                800 non-null    object
12  Cart_Abandonment_Factors                800 non-null    object
13  Saveforlater_Frequency                  800 non-null    object
14  Review_Left                             800 non-null    object
15  Review_Reliability                      800 non-null    object
16  Review_Helpfulness                      800 non-null    object
17  Personalized_Recommendation_Frequency  800 non-null    int64
18  Recommendation_Helpfulness              800 non-null    object
19  Rating_Accuracy                         800 non-null    int64
20  Shopping_Satisfaction                    800 non-null    int64
21  Service_Appreciation                    800 non-null    object
22  Improvement_Areas                       800 non-null    object
23  transaction                             800 non-null    int64
dtypes: int64(6), object(18)
memory usage: 150.1+ KB
```

```
SnapDeal.shape
```

```
(800, 24)
```

```
SnapDeal.columns
```

```
Index(['Timestamp', 'age', 'Gender', 'Purchase_Frequency',  
      'Purchase_Categories', 'Personalized_Recommendation_Frequency',  
      'Browsing_Frequency', 'Product_Search_Method',  
      'Search_Result_Exploration', 'Customer_Reviews_Importance',  
      'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',  
      'Cart_Abandonment_Factors', 'Saveforlater_Frequency',  
      'Review_Left',  
      'Review_Reliability', 'Review_Helpfulness',  
      'Personalized_Recommendation_Frequency',  
      'Recommendation_Helpfulness',  
      'Rating_Accuracy', 'Shopping_Satisfaction',  
      'Service_Appreciation',  
      'Improvement_Areas', 'transaction'],  
      dtype='object')
```

```
df.columns = df.columns.str.strip()  
print(df.columns.tolist())
```

```
['Timestamp', 'Age', 'Gender', 'Purchase_Frequency',  
 'Purchase_Categories', 'Personalized_Recommendation_Frequency',  
 'Browsing_Frequency', 'Product_Search_Method',  
 'Search_Result_Exploration', 'Customer_Reviews_Importance',  
 'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',  
 'Cart_Abandonment_Factors', 'Saveforlater_Frequency', 'Review_Left',  
 'Review_Reliability', 'Review_Helpfulness',  
 'Personalized_Recommendation_Frequency', 'Recommendation_Helpfulness',  
 'Rating_Accuracy', 'Shopping_Satisfaction', 'Service_Appreciation',  
 'Improvement_Areas', 'Transaction', 'Purchase_Frequency_Score']
```

```
SnapDeal.duplicated().sum()
```

```
0
```

```
SnapDeal.isnull().sum()
```

Timestamp	0
age	0
Gender	0
Purchase_Frequency	0
Purchase_Categories	0
Personalized_Recommendation_Frequency	0
Browsing_Frequency	0
Product_Search_Method	149
Search_Result_Exploration	0
Customer_Reviews_Importance	0

Add_to_Cart_Browsing	0
Cart_Completion_Frequency	0
Cart_Abandonment_Factors	0
Saveforlater_Frequency	0
Review_Left	0
Review_Reliability	0
Review_Helpfulness	0
Personalized_Recommendation_Frequency	0
Recommendation_Helpfulness	0
Rating_Accuracy	0
Shopping_Satisfaction	0
Service_Appreciation	0
Improvement_Areas	0
transaction	0

dtype: int64

SnapDeal.notnull().sum()

Timestamp	800
age	800
Gender	800
Purchase_Frequency	800
Purchase_Categories	800
Personalized_Recommendation_Frequency	800
Browsing_Frequency	800
Product_Search_Method	800
Search_Result_Exploration	800
Customer_Reviews_Importance	800
Add_to_Cart_Browsing	800
Cart_Completion_Frequency	800
Cart_Abandonment_Factors	800
Saveforlater_Frequency	800
Review_Left	800
Review_Reliability	800
Review_Helpfulness	800
Personalized_Recommendation_Frequency	800
Recommendation_Helpfulness	800
Rating_Accuracy	800
Shopping_Satisfaction	800
Service_Appreciation	800
Improvement_Areas	800
transaction	800

dtype: int64

```
# Replaced missing values in Product_Search_Method with "Others"
# Handle Missing Values
SnapDeal["Product_Search_Method"] =
(SnapDeal["Product_Search_Method"].replace(["", " ", "NA", "N/A"],
np.nan).fillna("Others"))
```

```
SnapDeal.head()
```

		Timestamp	age	Gender	\
0	2023/06/07 11:06:40	PM GMT+5:30	59	Others	
1	2023/06/06 7:08:47	PM GMT+5:30	48	Female	
2	2023/06/08 9:39:30	PM GMT+5:30	57	Prefer not to say	
3	2023/06/07 9:22:24	PM GMT+5:30	30	Others	
4	2023/06/06 7:21:26	PM GMT+5:30	51	Female	

	Purchase_Frequency	Purchase_Categories	\
0	Once a week	Beauty and Personal Care	
1	Once a month	Beauty and Personal Care;Clothing and Fashion;...	
2	Multiple times a week	Groceries and Gourmet Food	
3	Few times a month	Groceries and Gourmet Food;Beauty and Personal...	
4	Once a month	Groceries and Gourmet Food;Beauty and Personal...	

	Personalized_Recommendation_Frequency	Browsing_Frequency	\
0	Yes	Few times a month	
1	Sometimes	Rarely	
2	Sometimes	Rarely	
3	No	Few times a month	
4	No	Few times a month	

	Product_Search_Method	Search_Result_Exploration	\
0	others	Multiple pages	
1	categories	Multiple pages	
2	Keyword	First page	
3	others	Multiple pages	
4	Filter	First page	

	Customer_Reviews_Importance	...	Review_Left	Review_Reliability	\
0	4	...	Yes	Rarely	
1	1	...	No	Never	
2	2	...	Yes	Rarely	
3	2	...	Yes	Rarely	
4	5	...	Yes	Moderately	

	Review_Helpfulness	Personalized_Recommendation_Frequency	\
0	No	1	
1	Sometimes	1	
2	No	5	
3	Sometimes	4	
4	Sometimes	1	

	Recommendation_Helpfulness	Rating_Accuracy	Shopping_Satisfaction	\
0	Yes	3	3	
1	No	5	2	
2	Sometimes	3	3	
3	Sometimes	4	5	
4	Yes	3	1	

	Service_Appreciation	Improvement_Areas
transaction		
0	Customer service	User interface of app
433348		
1	.	.
735054		
2	Customer service	Product quality and accuracy
119350		
3	Customer service	.
180686		
4	.	I don't have any problem with Amazon
538184		

[5 rows x 24 columns]

SnapDeal.isnull().sum()

Timestamp	0
age	0
Gender	0
Purchase_Frequency	0
Purchase_Categories	0
Personalized_Recommendation_Frequency	0
Browsing_Frequency	0
Product_Search_Method	0
Search_Result_Exploration	0
Customer_Reviews_Importance	0
Add_to_Cart_Browsing	0
Cart_Completion_Frequency	0
Cart_Abandonment_Factors	0
Saveforlater_Frequency	0
Review_Left	0
Review_Reliability	0
Review_Helpfulness	0
Personalized_Recommendation_Frequency	0
Recommendation_Helpfulness	0
Rating_Accuracy	0
Shopping_Satisfaction	0
Service_Appreciation	0
Improvement_Areas	0
transaction	0
dtype: int64	

```
SnapDeal["Timestamp"] = pd.to_datetime(SnapDeal["Timestamp"],
errors="coerce")
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_18696\3931891890.py:1:
UserWarning: Could not infer format, so each element will be parsed
individually, falling back to `dateutil`. To ensure parsing is
consistent and as-expected, please specify a format.
```

```
SnapDeal["Timestamp"] = pd.to_datetime(SnapDeal["Timestamp"],
errors="coerce")
```

```
SnapDeal.dtypes
```

Timestamp	datetime64[ns, UTC-05:30]
age	int64
Gender	object
Purchase_Frequency	object
Purchase_Categories	object
Personalized_Recommendation_Frequency	object
Browsing_Frequency	object
Product_Search_Method	object
Search_Result_Exploration	object
Customer_Reviews_Importance	int64
Add_to_Cart_Browsing	object
Cart_Completion_Frequency	object
Cart_Abandonment_Factors	object
Saveforlater_Frequency	object
Review_Left	object
Review_Reliability	object
Review_Helpfulness	object
Personalized_Recommendation_Frequency	int64
Recommendation_Helpfulness	object
Rating_Accuracy	int64
Shopping_Satisfaction	int64
Service_Appreciation	object
Improvement_Areas	object
transaction	int64
dtype:	object

```
SnapDeal.shape
```

```
(800, 24)
```

```
SnapDeal.rename(columns={"age": "Age"}, inplace=True)
```

```
SnapDeal.rename(columns={"transaction": "Transaction"}, inplace=True)
```

```
SnapDeal.to_csv("SnapDeal_Cleaned.csv", index=False)
```

```
print("Cleaned dataset saved as 'SnapDeal_Cleaned.csv'")
```

```
Cleaned dataset saved as 'SnapDeal_Cleaned.csv'
```

```
df=pd.read_csv('SnapDeal_Cleaned.csv')
```

```
df.head()
```

	Timestamp	Age	Gender	\
0	2023/06/07 11:06:40 PM GMT+5:30	59	Others	
1	2023/06/06 7:08:47 PM GMT+5:30	48	Female	
2	2023/06/08 9:39:30 PM GMT+5:30	57	Prefer not to say	
3	2023/06/07 9:22:24 PM GMT+5:30	30	Others	
4	2023/06/06 7:21:26 PM GMT+5:30	51	Female	

	Purchase_Frequency	Purchase_Categories	\
0	Once a week	Beauty and Personal Care	
1	Once a month	Beauty and Personal Care;Clothing and Fashion;...	
2	Multiple times a week	Groceries and Gourmet Food	
3	Few times a month	Groceries and Gourmet Food;Beauty and Personal...	
4	Once a month	Groceries and Gourmet Food;Beauty and Personal...	

	Personalized_Recommendation_Frequency	Browsing_Frequency	\
0	Yes	Few times a month	
1	Sometimes	Rarely	
2	Sometimes	Rarely	
3	No	Few times a month	
4	No	Few times a month	

	Product_Search_Method	Search_Result_Exploration	\
0	others	Multiple pages	
1	categories	Multiple pages	
2	Keyword	First page	
3	others	Multiple pages	
4	Filter	First page	

	Customer_Reviews_Importance	... Review_Left	Review_Reliability	\
0	4	...	Yes	Rarely
1	1	...	No	Never
2	2	...	Yes	Rarely
3	2	...	Yes	Rarely
4	5	...	Yes	Moderately

	Review_Helpfulness	Personalized_Recommendation_Frequency	\
0	No	1	
1	Sometimes	1	
2	No	5	
3	Sometimes	4	

4	Sometimes	1
	Recommendation_Helpfulness	Rating_Accuracy
0	Yes	3
1	No	5
2	Sometimes	3
3	Sometimes	4
4	Yes	3

	Service_Appreciation	Improvement_Areas
Transaction		
0	Customer service	User interface of app
433348		
1	.	.
735054		
2	Customer service	Product quality and accuracy
119350		
3	Customer service	.
180686		
4	.	I don't have any problem with Amazon
538184		

[5 rows x 24 columns]

df.shape

(800, 24)

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 800 entries, 0 to 799

Data columns (total 24 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	Timestamp	800 non-null	object
1	Age	800 non-null	int64
2	Gender	800 non-null	object
3	Purchase_Frequency	800 non-null	object
4	Purchase_Categories	800 non-null	object
5	Personalized_Recommendation_Frequency	800 non-null	object
6	Browsing_Frequency	800 non-null	object
7	Product_Search_Method	800 non-null	object
8	Search_Result_Exploration	800 non-null	object
9	Customer_Reviews_Importance	800 non-null	int64
10	Add_to_Cart_Browsing	800 non-null	object
11	Cart_Completion_Frequency	800 non-null	object
12	Cart_Abandonment_Factors	800 non-null	object
13	Saveforlater_Frequency	800 non-null	object
14	Review_Left	800 non-null	object

15	Review_Reliability	800	non-null	object
16	Review_Helpfulness	800	non-null	object
17	Personalized_Recommendation_Frequency.1	800	non-null	int64
18	Recommendation_Helpfulness	800	non-null	object
19	Rating_Accuracy	800	non-null	int64
20	Shopping_Satisfaction	800	non-null	int64
21	Service_Appreciation	800	non-null	object
22	Improvement_Areas	800	non-null	object
23	Transaction	800	non-null	int64

dtypes: int64(6), object(18)

memory usage: 150.1+ KB

df.notnull().sum()

Timestamp	800
Age	800
Gender	800
Purchase_Frequency	800
Purchase_Categories	800
Personalized_Recommendation_Frequency	800
Browsing_Frequency	800
Product_Search_Method	800
Search_Result_Exploration	800
Customer_Reviews_Importance	800
Add_to_Cart_Browsing	800
Cart_Completion_Frequency	800
Cart_Abandonment_Factors	800
Saveforlater_Frequency	800
Review_Left	800
Review_Reliability	800
Review_Helpfulness	800
Personalized_Recommendation_Frequency.1	800
Recommendation_Helpfulness	800
Rating_Accuracy	800
Shopping_Satisfaction	800
Service_Appreciation	800
Improvement_Areas	800
Transaction	800

dtype: int64

df["Gender"]

0	Others
1	Female
2	Prefer Not To Say
3	Others
4	Female
...	
795	Male
796	Female

```
797      Female
798      Male
799      Others
Name: Gender, Length: 800, dtype: object
```

```
df["Customer_Reviews_Importance"]
```

```
0      4
1      1
2      2
3      2
4      5
..
795    5
796    3
797    5
798    2
799    2
Name: Customer_Reviews_Importance, Length: 800, dtype: int64
```

```
df["Age"]
```

```
0      59
1      48
2      57
3      30
4      51
..
795    60
796    56
797    48
798    40
799    67
Name: Age, Length: 800, dtype: int64
```

```
df["Shopping_Satisfaction"]
```

```
0      3
1      2
2      3
3      5
4      1
..
795    2
796    4
797    4
798    3
799    2
Name: Shopping_Satisfaction, Length: 800, dtype: int64
```

```
#Removed the duplicates or inconsistent survey responses.  
#Standardize categorical/numerical entries.  
#Handle missing values and inconsistent formats in  
Product_Search_Method and other fields.  
#Rename duplicate or misformatted columns.  
#Convert numerical rating columns and checked the datatypes.  
  
#Task 2-#Summarize customer demographics (age, gender distribution).
```

```
df["Age"].describe()
```

```
count      800.000000  
mean       36.053750  
std        19.310087  
min         3.000000  
25%        18.000000  
50%        37.000000  
75%        53.000000  
max        67.000000  
Name: Age, dtype: float64
```

```
df["Gender"].describe()
```

```
count      800  
unique       4  
top      Male  
freq      213  
Name: Gender, dtype: object
```

```
#Analyze overall purchase frequency and most popular product  
categories.
```

```
df["Purchase_Frequency"].value_counts()
```

```
Purchase_Frequency  
Multiple times a week      166  
Once a month              164  
Once a week               161  
Few times a month         157  
Less than once a month    152  
Name: count, dtype: int64
```

```
df["Purchase_Categories"].value_counts()
```

```
Purchase_Categories  
Clothing and Fashion      36  
Groceries and Gourmet Food;Beauty and Personal Care;Clothing and  
Fashion                   34  
Clothing and Fashion;Home and Kitchen  
34
```

Groceries and Gourmet Food;Beauty and Personal Care;Clothing and Fashion;Home and Kitchen 32
Groceries and Gourmet Food;Beauty and Personal Care;Clothing and Fashion;Home and Kitchen;others 32
Clothing and Fashion;others 32
Beauty and Personal Care;Clothing and Fashion;others 32
others 32
Groceries and Gourmet Food;Beauty and Personal Care;Home and Kitchen 32
Beauty and Personal Care;Clothing and Fashion;Home and Kitchen 31
Groceries and Gourmet Food;Clothing and Fashion;Home and Kitchen;others 31
Beauty and Personal Care;Clothing and Fashion 30
Beauty and Personal Care;others 29
Clothing and Fashion;Home and Kitchen;others 29
Groceries and Gourmet Food;Clothing and Fashion;others 28
Beauty and Personal Care;Clothing and Fashion;Home and Kitchen;others 28
Groceries and Gourmet Food 28
Home and Kitchen;others 26
Groceries and Gourmet Food;Beauty and Personal Care;others 26
Groceries and Gourmet Food;Beauty and Personal Care 25
Groceries and Gourmet Food;Beauty and Personal Care;Clothing and Fashion;others 25
Beauty and Personal Care;Home and Kitchen;others 24
Groceries and Gourmet Food;Home and Kitchen 24
Home and Kitchen 22
Groceries and Gourmet Food;Clothing and Fashion 22
Groceries and Gourmet Food;Home and Kitchen;others 22
Groceries and Gourmet Food;Clothing and Fashion;Home and Kitchen 20
Beauty and Personal Care;Home and Kitchen

```

18
Beauty and Personal Care
16
Name: count, dtype: int64

#most popular product categories are Clothing and fashion.

#Identify top browsing methods and most common cart abandonment factors.

df["Product_Search_Method"].value_counts()

Product_Search_Method
others      182
Keyword     163
categories  156
Filter      150
Others      149
Name: count, dtype: int64

df["Cart_Abandonment_Factors"].value_counts()

Cart_Abandonment_Factors
Found a better price elsewhere    219
others                          196
High shipping costs              194
Changed my mind or no longer need the item  191
Name: count, dtype: int64

#Calculate mean and median satisfaction, recommendation helpfulness, and rating accuracy.

df["Shopping_Satisfaction"].mean()

2.98875

df["Shopping_Satisfaction"].median()

3.0

df["Rating_Accuracy"].mean()

2.96375

df["Rating_Accuracy"].median()

3.0

#Generate summary statistics and visualizations for key behavioral variables.

behavior_columns = [
    "Shopping_Satisfaction",

```

```

    "Customer_Reviews_Importance",
    "Rating_Accuracy"
]

# Summary statistics
df[behavior_columns].describe()

```

	Shopping_Satisfaction	Customer_Reviews_Importance
Rating_Accuracy		
count	800.000000	800.000000
800.000000		
mean	2.988750	2.988750
2.96375		
std	1.407515	1.400384
1.42565		
min	1.000000	1.000000
1.00000		
25%	2.000000	2.000000
2.00000		
50%	3.000000	3.000000
3.00000		
75%	4.000000	4.000000
4.00000		
max	5.000000	5.000000
5.00000		

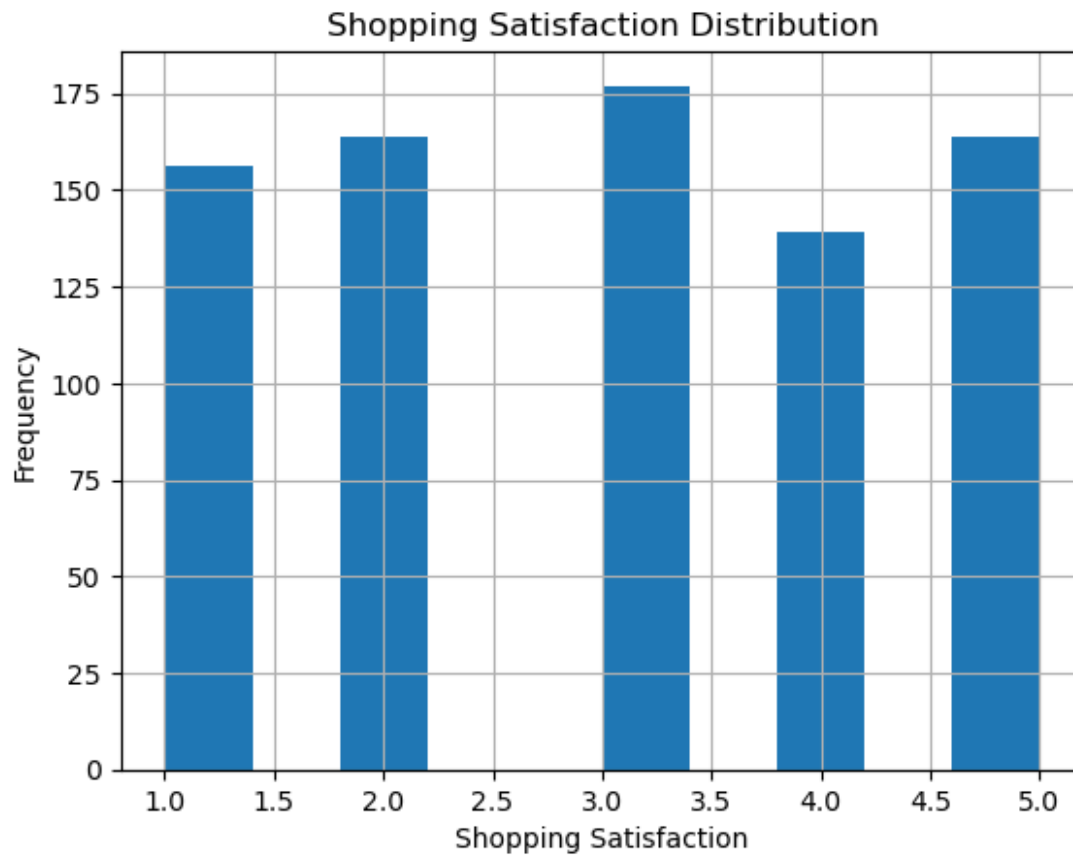
```

df["Shopping_Satisfaction"].hist(bins=10)

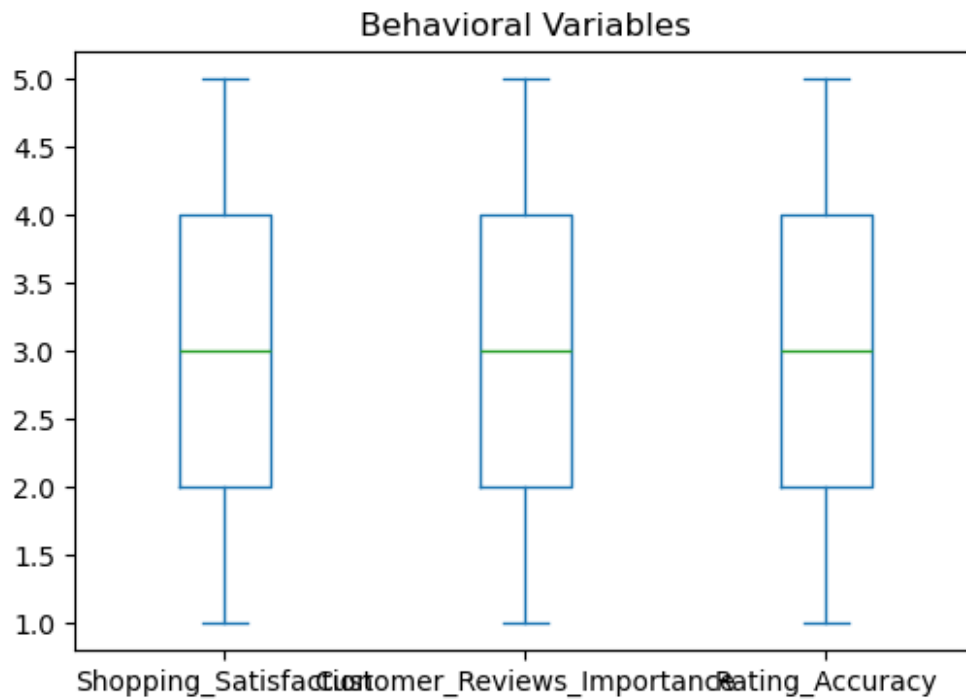
plt.title("Shopping Satisfaction Distribution")
plt.xlabel("Shopping Satisfaction")
plt.ylabel("Frequency")

plt.show()

```



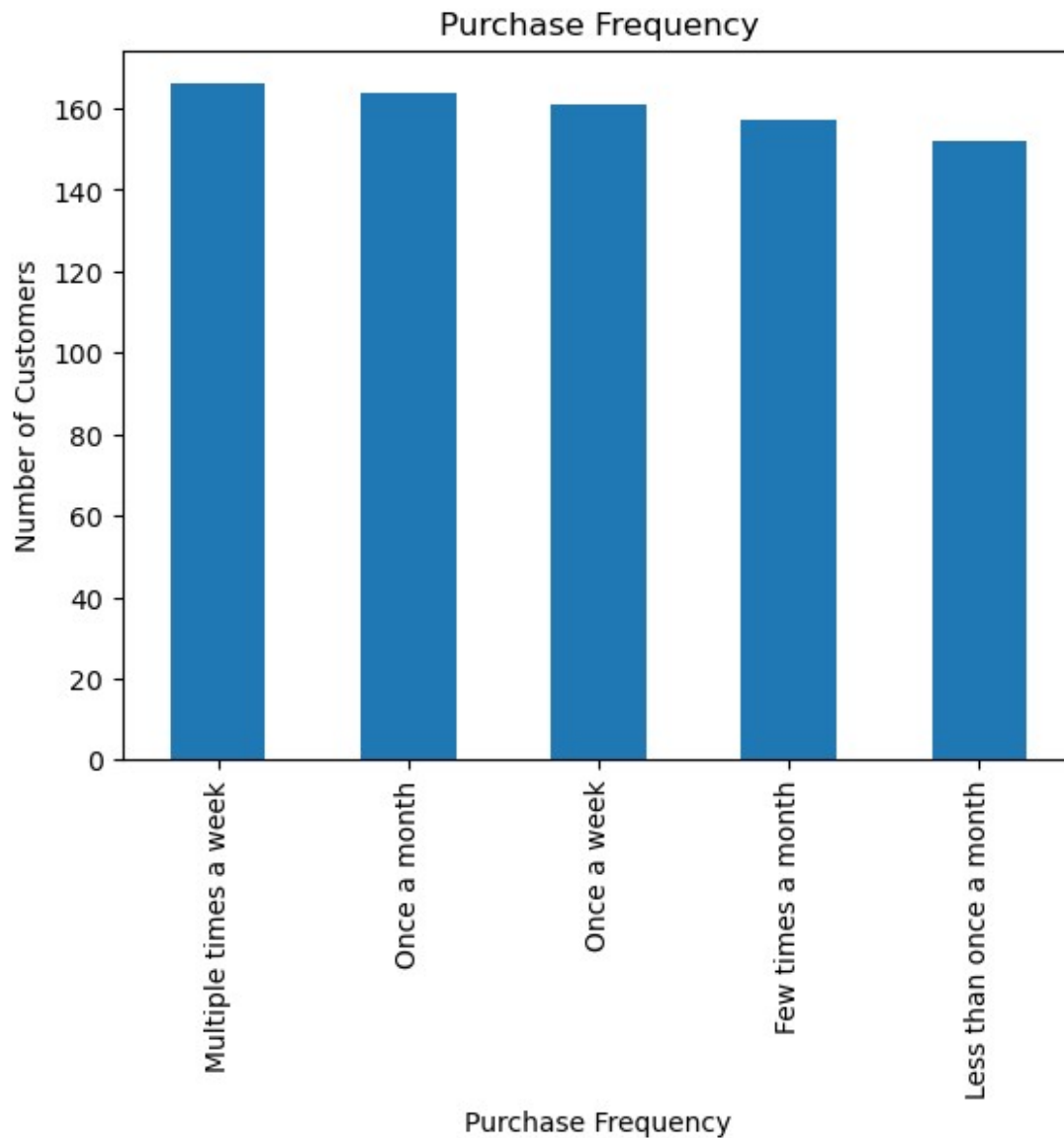
```
df[behavior_columns].plot(kind="box", figsize=(6,4))  
plt.title("Behavioral Variables")  
plt.show()
```



```
df["Purchase_Frequency"].value_counts().plot(
    kind="bar"
)

plt.title("Purchase Frequency")
plt.xlabel("Purchase Frequency")
plt.ylabel("Number of Customers")

plt.show()
```

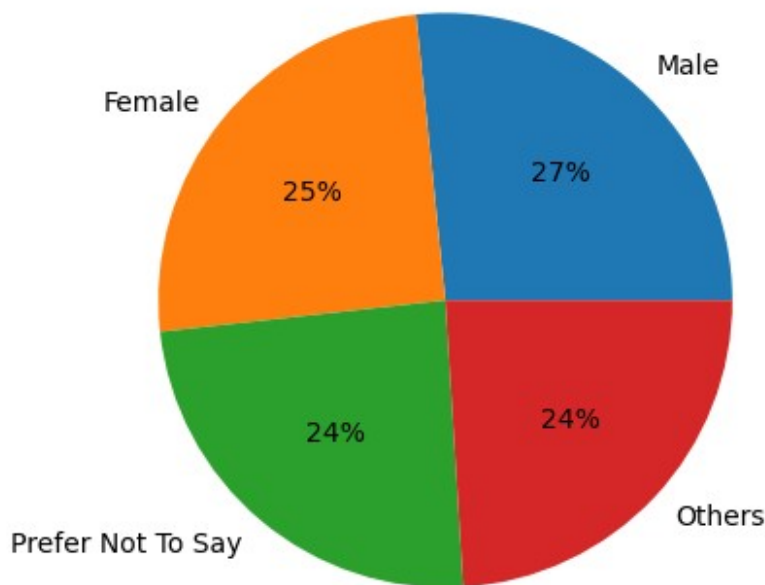



```
df["Gender"].value_counts().plot(
    kind="pie",
    autopct="%1.0f%%"
)

plt.title("Gender Distribution")
plt.ylabel("")

plt.show()
```

Gender Distribution



#Task 3: Customer Segmentation and Profiling (20 Marks)

#● Segment customers based on purchase frequency and shopping satisfaction levels.

#● Create profiles such as:

#○ Frequent Buyers: High purchase frequency, high satisfaction.

#○ Occasional Shoppers: Medium frequency, moderate satisfaction.

#○ At-Risk Customers: Low satisfaction or frequent cart abandonment.

#● Analyze demographic or behavioral differences across these segments.

#● Use clustering (e.g., K-Means) for behavioral grouping based on survey responses.

`df.head()`

	Timestamp	age	Gender \
0	2023/06/07 11:06:40 PM GMT+5:30	59	Others
1	2023/06/06 7:08:47 PM GMT+5:30	48	Female
2	2023/06/08 9:39:30 PM GMT+5:30	57	Prefer not to say
3	2023/06/07 9:22:24 PM GMT+5:30	30	Others
4	2023/06/06 7:21:26 PM GMT+5:30	51	Female

Purchase_Frequency	Purchase_Categories \
0	Once a week
	Beauty and Personal Care

1	Once a month	Beauty and Personal Care;Clothing and Fashion;...
2	Multiple times a week	Groceries and Gourmet Food
3	Few times a month	Groceries and Gourmet Food;Beauty and Personal...
4	Once a month	Groceries and Gourmet Food;Beauty and Personal...

	Personalized_Recommendation_Frequency	Browsing_Frequency	\
0	Yes	Few times a month	
1	Sometimes	Rarely	
2	Sometimes	Rarely	
3	No	Few times a month	
4	No	Few times a month	

	Product_Search_Method	Search_Result_Exploration	\
0	others	Multiple pages	
1	categories	Multiple pages	
2	Keyword	First page	
3	others	Multiple pages	
4	Filter	First page	

	Customer_Reviews_Importance	...	Review_Left	Review_Reliability	\
0	4	...	Yes	Rarely	
1	1	...	No	Never	
2	2	...	Yes	Rarely	
3	2	...	Yes	Rarely	
4	5	...	Yes	Moderately	

	Review_Helpfulness	Personalized_Recommendation_Frequency	\
0	No	1	
1	Sometimes	1	
2	No	5	
3	Sometimes	4	
4	Sometimes	1	

	Recommendation_Helpfulness	Rating_Accuracy	Shopping_Satisfaction	\
0	Yes	3	3	
1	No	5	2	
2	Sometimes	3	3	
3	Sometimes	4	5	
4	Yes	3	1	

	Service_Appreciation	Improvement_Areas
transaction		
0	Customer service	User interface of app
433348		
1	.	.
735054		

```

2      Customer service      Product quality and accuracy
119350
3      Customer service      .
180686
4      . I don't have any problem with Amazon
538184

```

```
[5 rows x 24 columns]
```

```
df["Purchase_Frequency"]
```

```

0      Once a week
1      Once a month
2      Multiple times a week
3      Few times a month
4      Once a month

```

```
...
```

```

795    Less than once a month
796    Multiple times a week
797    Multiple times a week
798    Multiple times a week
799    Multiple times a week

```

```
Name: Purchase_Frequency, Length: 800, dtype: object
```

```
df.groupby("Purchase_Frequency").size()
```

```

Purchase_Frequency
Few times a month      157
Less than once a month 152
Multiple times a week  166
Once a month           164
Once a week            161
dtype: int64

```

```
def customer_segment(row):
```

```

    if row["Purchase_Frequency"] in ["Once a week", "Multiple times a
week"] and row["Shopping_Satisfaction"] >= 4:
        return "Frequent Buyers"

```

```

    elif row["Purchase_Frequency"] in ["Few times a month", "Once a
month"] and row["Shopping_Satisfaction"] >= 3:
        return "Occasional Shoppers"

```

```

    elif row["Purchase_Frequency"] == "Less than once a month" or
row["Shopping_Satisfaction"] <= 2:
        return "At-Risk Customers"

```

```

    else:
        return "Other"

```

```
df["Customer_Segment"] = df.apply(customer_segment, axis=1)
segment_counts = df["Customer_Segment"].value_counts()
print(segment_counts)
```

```
Customer_Segment
At-Risk Customers      421
Occasional Shoppers    183
Frequent Buyers        128
Other                   68
Name: count, dtype: int64
```

```
grouped = df.groupby("Purchase_Frequency")
for frequency, group in grouped:
    display(group)
```

	Timestamp	age	Gender \
3	2023/06/07 9:22:24 PM GMT+5:30	30	Others
14	2023/06/09 8:10:05 AM GMT+5:30	15	Female
24	2023/06/09 10:23:05 AM GMT+5:30	54	Male
28	2023/06/08 8:26:58 PM GMT+5:30	20	Female
42	2023/06/08 10:24:19 PM GMT+5:30	48	Male
...	...	...	...
772	2023/06/08 5:07:35 PM GMT+5:30	55	Prefer not to say
777	2023/06/08 5:43:45 PM GMT+5:30	40	Others
781	2023/06/09 10:28:57 AM GMT+5:30	8	Others
785	2023/06/05 12:05:20 AM GMT+5:30	60	Female
788	2023/06/11 10:43:19 PM GMT+5:30	47	Others

Purchase_Frequency	Purchase_Categories \
3	Few times a month Groceries and Gourmet Food;Beauty and Personal...
14	Few times a month Beauty and Personal Care;Clothing and Fashion
24	Few times a month Home and Kitchen;others
28	Few times a month Clothing and Fashion;Home and Kitchen;others
42	Few times a month Beauty and Personal Care;others
...	...
...	...
772	Few times a month Beauty and Personal Care;Clothing and Fashion
777	Few times a month Clothing and Fashion;Home and Kitchen
781	Few times a month Beauty and Personal Care;Clothing and

Fashion;...

785 Few times a month Groceries and Gourmet Food;Beauty and Personal...

788 Few times a month Beauty and Personal Care;Clothing and Fashion

	Personalized_Recommendation_Frequency	Browsing_Frequency \
3	No	Few times a month
14	Sometimes	Few times a week
24	Yes	Few times a month
28	Sometimes	Few times a month
42	No	Rarely
..	...	...
772	No	Few times a month
777	Yes	Rarely
781	Sometimes	Few times a month
785	Sometimes	Multiple times a day
788	Sometimes	Few times a week

	Product_Search_Method	Search_Result_Exploration \
3	others	Multiple pages
14	categories	First page
24	Keyword	First page
28	NaN	Multiple pages
42	Keyword	Multiple pages
..	...	...
772	Keyword	First page
777	others	First page
781	Keyword	First page
785	others	Multiple pages
788	Keyword	Multiple pages

	Customer_Reviews_Importance	... Review_Reliability
Review_Helpfulness \		
3	2 ...	Rarely
Sometimes		
14	4 ...	Never
Sometimes		
24	1 ...	Moderately
No		
28	1 ...	Occasionally
No		
42	1 ...	Rarely
Yes		
..		...
...		
772	1 ...	Rarely
Sometimes		
777	2 ...	Never
Sometimes		

781	3	...	Never
No			
785	5	...	Moderately
No			
788	4	...	Never
Yes			
Personalized_Recommendation_Frequency Recommendation_Helpfulness			
\			
3		4	Sometimes
14		3	No
24		2	Sometimes
28		5	Sometimes
42		4	Sometimes
..		...	...
772		4	No
777		5	Yes
781		3	No
785		1	No
788		3	Yes
Rating_Accuracy Shopping_Satisfaction Service_Appreciation \			
3	4	5	Customer service
14	4	5	Customer service
24	4	4	.
28	5	3	Customer service
42	2	1	Wide product selection
..	...	...	...
772	5	1	Competitive prices
777	5	2	Competitive prices
781	5	1	Customer service
785	2	3	Competitive prices
788	3	3	Product recommendations
Improvement_Areas transaction \			
3		.	180686
14	I have no problem with Amazon yet. But others	...	397056
24		.	738929
28		UI	742772
42	No problems with Amazon		922302

772	Quality of product is very poor according to t...	850593
777	Reducing packaging waste	880702
781	I don't have any problem with Amazon	708679
785	Nil	190661
788	Product quality and accuracy	102172

	Customer_Segment
3	Occasional Shoppers
14	Occasional Shoppers
24	Occasional Shoppers
28	Occasional Shoppers
42	At-Risk Customers
772	At-Risk Customers
777	At-Risk Customers
781	At-Risk Customers
785	Occasional Shoppers
788	Occasional Shoppers

[157 rows x 25 columns]

	Timestamp	age	Gender \
10	2023/06/07 10:10:58 PM GMT+5:30	28	Others
18	2023/06/09 2:51:07 PM GMT+5:30	51	Prefer not to say
20	2023/06/08 3:23:58 AM GMT+5:30	28	Male
21	2023/06/06 8:56:17 PM GMT+5:30	10	Prefer not to say
22	2023/06/11 11:15:03 PM GMT+5:30	11	Female
776	2023/06/06 7:18:01 PM GMT+5:30	17	Male
782	2023/06/08 9:40:24 PM GMT+5:30	51	Prefer not to say
789	2023/06/08 3:29:26 AM GMT+5:30	47	Female
793	2023/06/09 9:37:44 AM GMT+5:30	4	Male
795	2023/06/11 2:07:39 PM GMT+5:30	60	Male

	Purchase_Frequency \
10	Less than once a month
18	Less than once a month
20	Less than once a month
21	Less than once a month
22	Less than once a month
776	Less than once a month
782	Less than once a month
789	Less than once a month
793	Less than once a month
795	Less than once a month

	Purchase_Categories \
10	Beauty and Personal Care;Clothing and Fashion;...

18	Groceries and Gourmet Food;Beauty and Personal...
20	Groceries and Gourmet Food;Beauty and Personal...
21	Groceries and Gourmet Food;Beauty and Personal...
22	Groceries and Gourmet Food;Beauty and Personal...
..	...
776	Groceries and Gourmet Food;Clothing and Fashion
782	Groceries and Gourmet Food;Beauty and Personal...
789	Clothing and Fashion
793	Clothing and Fashion;Home and Kitchen;others
795	Groceries and Gourmet Food;Beauty and Personal...

	Personalized_Recommendation_Frequency	Browsing_Frequency \
10	Sometimes	Multiple times a day
18	No	Few times a month
20	No	Few times a month
21	Sometimes	Few times a week
22	No	Rarely
..	...	...
776	No	Few times a month
782	Yes	Few times a week
789	Yes	Multiple times a day
793	Sometimes	Few times a week
795	Sometimes	Multiple times a day

	Product_Search_Method	Search_Result_Exploration \
10	Keyword	First page
18	Keyword	First page
20	categories	First page
21	others	Multiple pages
22	others	Multiple pages
..	...	...
776	NaN	Multiple pages
782	Keyword	Multiple pages
789	NaN	First page
793	others	Multiple pages
795	others	First page

	Customer_Reviews_Importance	... Review_Reliability
Review_Helpfulness \		
10	5 ...	Rarely
Sometimes		
18	5 ...	Rarely
No		
20	3 ...	Rarely
Yes		
21	5 ...	Never
Sometimes		
22	5 ...	Moderately
No		
..		...

...			
776	1	...	Rarely
Sometimes			
782	4	...	Occasionally
No			
789	5	...	Heavily
Sometimes			
793	2	...	Moderately
No			
795	5	...	Rarely
Sometimes			

	Personalized_Recommendation_Frequency	Recommendation_Helpfulness
\		
10	5	Sometimes
18	1	Sometimes
20	1	No
21	2	Sometimes
22	5	No
..	...	...
776	1	Sometimes
782	3	Sometimes
789	2	No
793	5	Yes
795	3	Sometimes

	Rating_Accuracy	Shopping_Satisfaction	\
10	4	4	
18	3	3	
20	2	4	
21	5	2	
22	4	4	
..	...	...	
776	5	3	
782	3	3	
789	4	4	
793	1	3	
795	4	2	

Service_Appreciation \

10	Customer service
18	Quick delivery
20	All the above
21	Customer service
22	.
..	...
776	Wide product selection
782	Customer service
789	User-friendly website/app interface
793	User-friendly website/app interface
795	Customer service

	Improvement_Areas	transaction \
10	Customer service responsiveness	797322
18	I don't have any problem with Amazon	119213
20	Customer service responsiveness	133335
21	Shipping speed and reliability	886529
22	I don't have any problem with Amazon	362553
..	...	...
776	Scrolling option would be much better than goi...	890872
782	User interface of app	751973
789	I don't have any problem with Amazon	421355
793	Reducing packaging waste	537502
795	Nothing	526649

	Customer_Segment
10	At-Risk Customers
18	At-Risk Customers
20	At-Risk Customers
21	At-Risk Customers
22	At-Risk Customers
..	...
776	At-Risk Customers
782	At-Risk Customers
789	At-Risk Customers
793	At-Risk Customers
795	At-Risk Customers

[152 rows x 25 columns]

	Timestamp	age	Gender \
2	2023/06/08 9:39:30 PM GMT+5:30	57	Prefer not to say
5	2023/06/04 11:27:53 PM GMT+5:30	33	Female
12	2023/06/09 9:33:35 AM GMT+5:30	7	Others
13	2023/06/08 5:15:27 PM GMT+5:30	48	Male
19	2023/06/08 5:35:01 PM GMT+5:30	56	Male
..	...	...	...
794	2023/06/07 12:13:39 PM GMT+5:30	19	Male
796	2023/06/09 7:22:11 PM GMT+5:30	56	Female
797	2023/06/12 3:47:01 PM GMT+5:30	48	Female

798	2023/06/06 6:51:13 PM GMT+5:30	40	Male
799	2023/06/09 10:26:51 AM GMT+5:30	67	Others

Purchase_Frequency	Purchase_Categories \
2	Multiple times a week Groceries and Gourmet Food
5	Multiple times a week Groceries and Gourmet Food;Beauty and Personal...
12	Multiple times a week others
13	Multiple times a week Groceries and Gourmet Food;Beauty and Personal...
19	Multiple times a week Clothing and Fashion;Home and Kitchen;others
..	...
...	
794	Multiple times a week Groceries and Gourmet Food;Clothing and Fashion...
796	Multiple times a week Beauty and Personal Care;Clothing and Fashion;...
797	Multiple times a week Groceries and Gourmet Food
798	Multiple times a week Groceries and Gourmet Food;Beauty and Personal...
799	Multiple times a week Clothing and Fashion;Home and Kitchen

Personalized_Recommendation_Frequency	Browsing_Frequency \
2	Sometimes Rarely
5	Yes Few times a month
12	Sometimes Multiple times a day
13	Yes Few times a week
19	Sometimes Few times a week
..	...
794	Sometimes Few times a month
796	Yes Rarely
797	Sometimes Multiple times a day
798	No Few times a month
799	Sometimes Multiple times a day

Product_Search_Method	Search_Result_Exploration \
2	Keyword First page
5	categories Multiple pages
12	NaN First page
13	others First page
19	Keyword Multiple pages
..	...
794	Keyword Multiple pages
796	NaN Multiple pages

797	others	Multiple pages
798	Keyword	Multiple pages
799	Keyword	First page

Customer_Reviews_Importance	...	Review_Reliability
Review_Helpfulness \		
2	2 ...	Rarely
No		
5	3 ...	Occasionally
Sometimes		
12	3 ...	Rarely
Sometimes		
13	5 ...	Heavily
Sometimes		
19	3 ...	Moderately
No		
..		...
...		
794	3 ...	Moderately
Sometimes		
796	3 ...	Occasionally
Yes		
797	5 ...	Heavily
Yes		
798	2 ...	Occasionally
No		
799	2 ...	Occasionally
Sometimes		

Personalized_Recommendation_Frequency	Recommendation_Helpfulness
\	
2	5 Sometimes
5	5 Sometimes
12	5 Yes
13	3 Sometimes
19	3 Sometimes
..	
794	4 Sometimes
796	4 Sometimes
797	5 Yes
798	4 Sometimes

799	1	Sometimes
-----	---	-----------

	Rating_Accuracy	Shopping_Satisfaction	Service_Appreciation \
2	3	3	Customer service
5	4	4	Customer service
12	5	5	Quick delivery
13	5	5	Competitive prices
19	3	3	.
..	...	...	...
794	4	1	Competitive prices
796	4	4	Customer service
797	2	4	Wide product selection
798	4	3	Quick delivery
799	3	2	All the above

Improvement_Areas transaction			
Customer_Segment			
2	Product quality and accuracy	119350	
Other			
5	Customer service responsiveness	116480	Frequent
Buyers			
12	Shipping speed and reliability	785442	Frequent
Buyers			
13	Nothing	769031	Frequent
Buyers			
19	No problems with Amazon	774345	
Other			
..	...	...	.
..			
794	Add more familiar brands to the list	312813	At-Risk
Customers			
796	.	524319	Frequent
Buyers			
797	Product quality and accuracy	637762	Frequent
Buyers			
798	Nil	290410	
Other			
799	Irrelevant product suggestions	867703	At-Risk
Customers			

[166 rows x 25 columns]

	Timestamp	age	Gender \
1	2023/06/06 7:08:47 PM GMT+5:30	48	Female
4	2023/06/06 7:21:26 PM GMT+5:30	51	Female
6	2023/06/06 7:40:53 PM GMT+5:30	10	Male
8	2023/06/08 9:39:55 PM GMT+5:30	8	Female
11	2023/06/09 9:35:57 AM GMT+5:30	15	Others

780	2023/06/07 6:30:43 PM GMT+5:30	16	Prefer not to say
784	2023/06/04 6:31:41 PM GMT+5:30	29	Male
787	2023/06/07 9:32:38 AM GMT+5:30	43	Others
790	2023/06/06 7:10:14 PM GMT+5:30	46	Others
792	2023/06/09 7:24:06 PM GMT+5:30	64	Male

Purchase_Frequency		
Purchase_Categories \		
1	Once a month	Beauty and Personal Care;Clothing and Fashion;...
4	Once a month	Groceries and Gourmet Food;Beauty and Personal...
6	Once a month	Groceries and Gourmet Food;Clothing and Fashio...
8	Once a month	Clothing and Fashion
11	Once a month	Clothing and Fashion;Home and Kitchen
..	...	
...		
780	Once a month	Groceries and Gourmet Food
784	Once a month	Clothing and Fashion;Home and Kitchen
787	Once a month	Beauty and Personal Care;Clothing and Fashion;...
790	Once a month	Groceries and Gourmet Food;Home and Kitchen;ot...
792	Once a month	Groceries and Gourmet Food;Home and Kitchen

Personalized_Recommendation_Frequency	Browsing_Frequency \
1	Sometimes Rarely
4	No Few times a month
6	Sometimes Few times a week
8	Sometimes Few times a week
11	Yes Few times a month
..	...
780	Yes Few times a week
784	Yes Rarely
787	Sometimes Few times a week
790	Sometimes Multiple times a day
792	Yes Multiple times a day

Product_Search_Method	Search_Result_Exploration \
1	categories Multiple pages
4	Filter First page
6	NaN Multiple pages
8	Filter First page

11	NaN	First page
...	...	...
780	NaN	Multiple pages
784	Filter	First page
787	Filter	Multiple pages
790	Keyword	First page
792	Filter	First page

Customer_Reviews_Importance	...	Review_Reliability
Review_Helpfulness \		
1	1 ...	Never
Sometimes		
4	5 ...	Moderately
Sometimes		
6	5 ...	Rarely
Sometimes		
8	3 ...	Never
Yes		
11	3 ...	Never
No		
...		...
...		
780	2 ...	Never
No		
784	1 ...	Heavily
Yes		
787	1 ...	Heavily
Yes		
790	5 ...	Moderately
Sometimes		
792	4 ...	Never
Yes		

Personalized_Recommendation_Frequency	Recommendation_Helpfulness
\	
1	1 No
4	1 Yes
6	1 Yes
8	5 Sometimes
11	4 Yes
...	
780	5 Yes
784	5 No

787	2	No
790	3	No
792	4	No

	Rating_Accuracy	Shopping_Satisfaction	\
1	5	2	
4	3	1	
6	3	2	
8	1	2	
11	5	3	
..	...	...	
780	2	2	
784	5	2	
787	4	2	
790	4	1	
792	2	4	

	Service_Appreciation	\
1	.	
4	.	
6	Wide product selection	
8	All the above	
11	User-friendly website/app interface	
..	...	
780	Customer service	
784	User-friendly website/app interface	
787	Customer service	
790	All the above	
792	Quick delivery	

	Improvement_Areas	transaction	\
1	.	735054	
4	I don't have any problem with Amazon	538184	
6	Nil	445746	
8	No problems with Amazon	571022	
11	better app interface and lower shipping charges	252740	
..	...	...	
780	I have no problem with Amazon yet. But others	834597	
784	Add more familiar brands to the list	299929	
787	User interface	668260	
790	Irrelevant product suggestions	720260	
792	better app interface and lower shipping charges	545801	

	Customer_Segment
1	At-Risk Customers
4	At-Risk Customers

```

6      At-Risk Customers
8      At-Risk Customers
11     Occasional Shoppers
..
780    At-Risk Customers
784    At-Risk Customers
787    At-Risk Customers
790    At-Risk Customers
792    Occasional Shoppers

```

[164 rows x 25 columns]

```

                                Timestamp age      Gender \
0      2023/06/07 11:06:40 PM GMT+5:30  59      Others
7      2023/06/06 5:16:58 PM GMT+5:30  30      Male
9      2023/06/11 11:10:14 PM GMT+5:30  11      Female
15     2023/06/07 11:41:56 AM GMT+5:30  58      Male
17     2023/06/04 8:00:11 PM GMT+5:30  60      Others
..
728    2023/06/08 3:24:37 AM GMT+5:30  28      Prefer not to say
759    2023/06/07 9:18:55 PM GMT+5:30  63      Prefer not to say
767    2023/06/07 11:41:56 AM GMT+5:30  51      Female
770    2023/06/07 1:25:26 PM GMT+5:30  40      Prefer not to say
786    2023/06/09 3:24:36 PM GMT+5:30  22      Others

```

```

Purchase_Frequency
Purchase_Categories \
0      Once a week      Beauty and Personal
Care
7      Once a week      Groceries and Gourmet Food;Beauty and
Personal...
9      Once a week      Clothing and
Fashion
15     Once a week      Groceries and Gourmet Food;Clothing and
Fashio...
17     Once a week      Clothing and Fashion;Home and
Kitchen;others
..
728    Once a week      Beauty and Personal Care;Clothing and
Fashion;...
759    Once a week      Groceries and Gourmet Food;Beauty and
Personal...
767    Once a week      Beauty and Personal Care;Clothing and
Fashion;...
770    Once a week      Groceries and Gourmet Food;Home and
Kitchen
786    Once a week      Home and
Kitchen

```

Personalized_Recommendation_Frequency		Browsing_Frequency \	
0	Yes	Few times a month	
7	No	Few times a month	
9	Yes	Few times a month	
15	No	Rarely	
17	No	Few times a week	
..	...	...	
728	Yes	Rarely	
759	Sometimes	Few times a week	
767	Sometimes	Rarely	
770	Yes	Multiple times a day	
786	Sometimes	Few times a month	
Product_Search_Method		Search_Result_Exploration \	
0	others	Multiple pages	
7	categories	First page	
9	categories	Multiple pages	
15	categories	Multiple pages	
17	others	First page	
..	...	...	
728	categories	First page	
759	Filter	First page	
767	categories	Multiple pages	
770	NaN	Multiple pages	
786	others	Multiple pages	
Customer_Reviews_Importance		... Review_Reliability	
Review_Helpfulness \			
0	4	...	Rarely
No			
7	2	...	Occasionally
Sometimes			
9	3	...	Heavily
No			
15	4	...	Rarely
Sometimes			
17	4	...	Never
No			
..	...	...	...
...			
728	3	...	Occasionally
No			
759	5	...	Heavily
No			
767	5	...	Never
Sometimes			
770	4	...	Moderately
Sometimes			
786	2	...	Never

Yes

	Personalized_Recommendation_Frequency	Recommendation_Helpfulness
0	1	Yes
7	1	Sometimes
9	3	No
15	3	No
17	4	Sometimes
..	...	...
728	5	No
759	2	No
767	1	Yes
770	4	Yes
786	3	No

	Rating_Accuracy	Shopping_Satisfaction	Service_Appreciation	\
0	3	3	Customer service	
7	5	2	Wide product selection	
9	2	3	Customer service	
15	4	2	Product recommendations	
17	3	4	Customer service	
..	...	...	...	
728	5	2	Product recommendations	
759	2	5	Competitive prices	
767	3	1	Customer service	
770	3	2	Wide product selection	
786	2	4	Competitive prices	

	Improvement_Areas	transaction	\
0	User interface of app	433348	
7	No problems with Amazon	372797	
9	Irrelevant product suggestions	952835	
15	Shipping speed and reliability	757928	
17	I have no problem with Amazon yet. But others ...	701440	
..	...	...	
728	Shipping speed and reliability	745511	
759	Reducing packaging waste	565915	
767	No problems with Amazon	818624	
770	User interface of app	837191	

786 Irrelevant product suggestions 556093

```
Customer_Segment
0 Other
7 At-Risk Customers
9 Other
15 At-Risk Customers
17 Frequent Buyers
..
728 At-Risk Customers
759 Frequent Buyers
767 At-Risk Customers
770 At-Risk Customers
786 Frequent Buyers
```

[161 rows x 25 columns]

#Above value is Customer segmentation, based on the purchase frequency.

#profiles such as:

#○ Frequent Buyers: High purchase frequency, high satisfaction.

#○ Occasional Shoppers: Medium frequency, moderate satisfaction.

#○ At-Risk Customers: Low satisfaction or frequent cart abandonment.

```
def customer_profile(row):
```

```
    # Frequent Buyers
```

```
    if (row["Purchase_Frequency"] in ["Multiple times a week", "Once a week"]) and
```

```
        row["Shopping_Satisfaction"] >= 4):
```

```
        return "Frequent Buyers"
```

```
    # Occasional Shoppers
```

```
    elif (row["Purchase_Frequency"] in ["Few times a month", "Once a month"]) and
```

```
        row["Shopping_Satisfaction"] >= 3):
```

```
        return "Occasional Shoppers"
```

```
    # At-Risk Customers
```

```
    elif (row["Shopping_Satisfaction"] <= 2 or
```

```
        row["Cart_Abandonment_Factors"] != "None"):
```

```
        return "At-Risk Customers"
```

```
    else:
```

```
        return "Other"
```

```
df["Customer_Profile"] = df.apply(customer_profile, axis=1)
```

```
df["Customer_Profile"].value_counts()
```

```
Customer_Profile
At-Risk Customers      489
Occasional Shoppers    183
Frequent Buyers        128
Name: count, dtype: int64
```

```
#Frequent Buyers: High purchase frequency, high satisfaction.=128
#Occasional Shoppers: Medium frequency, moderate satisfaction.=183
#At-Risk Customers: Low satisfaction or frequent cart abandonment.=489
```

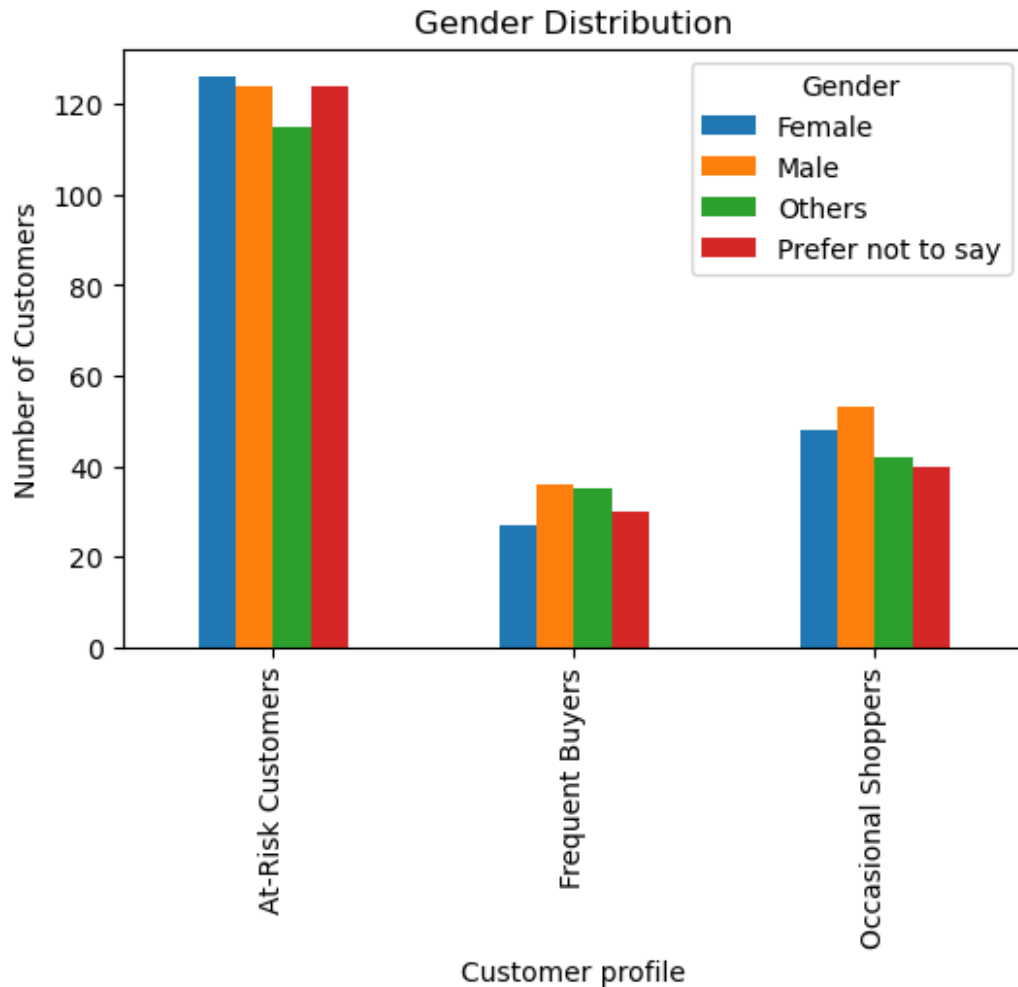
```
#Analyze demographic or behavioral differences across these segments.
```

```
#Demographic analysis across Gender.
```

```
gender_profile = pd.crosstab(df["Customer_Profile"], df["Gender"])
gender_profile
```

Gender	Female	Male	Others	Prefer not to say
Customer_Profile				
At-Risk Customers	126	124	115	124
Frequent Buyers	27	36	35	30
Occasional Shoppers	48	53	42	40

```
gender_profile.plot(kind="bar", figsize=(6,4))
plt.title("Gender Distribution")
plt.xlabel("Customer profile")
plt.ylabel("Number of Customers")
plt.show()
```



#Behavioral Analysis

```
df.groupby("Customer_Profile")["Shopping_Satisfaction"].mean()
```

```
Customer_Profile
At-Risk Customers    2.226994
Frequent Buyers      4.562500
Occasional Shoppers  3.923497
Name: Shopping_Satisfaction, dtype: float64
```

```
pd.crosstab(df["Customer_Profile"], df["Purchase_Frequency"])
```

```
Purchase_Frequency  Few times a month  Less than once a month \
Customer_Profile
At-Risk Customers           67             152
Frequent Buyers              0              0
Occasional Shoppers         90              0

Purchase_Frequency  Multiple times a week  Once a month  Once a week
```

Customer_Profile			
At-Risk Customers	104	71	95
Frequent Buyers	62	0	66
Occasional Shoppers	0	93	0

```
pd.crosstab(
    df["Customer_Profile"],
    df["Cart_Abandonment_Factors"]
)
```

Cart_Abandonment_Factors Changed my mind or no longer need the item \

Customer_Profile

At-Risk Customers	119
Frequent Buyers	33
Occasional Shoppers	39

Cart_Abandonment_Factors Found a better price elsewhere High shipping costs \

Customer_Profile

At-Risk Customers	135
108	
Frequent Buyers	39
38	
Occasional Shoppers	45
48	

Cart_Abandonment_Factors others

Customer_Profile

At-Risk Customers	127
Frequent Buyers	18
Occasional Shoppers	51

#Above Behavioral Analysis on the basis of Shopping_Satisfaction,Purchase_Frequency,Cart_Abandonment_Factors.

#Use clustering (e.g., K-Means) for behavioral grouping based on survey responses.

```
df.head(3)
```

	Timestamp	age	Gender \
0	2023/06/07 11:06:40 PM GMT+5:30	59	Others

1	2023/06/06 7:08:47 PM GMT+5:30	48	Female
2	2023/06/08 9:39:30 PM GMT+5:30	57	Prefer not to say

Purchase_Frequency		Purchase_Categories \
0	Once a week	Beauty and Personal Care
1	Once a month	Beauty and Personal Care;Clothing and Fashion;...
2	Multiple times a week	Groceries and Gourmet Food

Personalized_Recommendation_Frequency		Browsing_Frequency \
0	Yes	Few times a month
1	Sometimes	Rarely
2	Sometimes	Rarely

Product_Search_Method		Search_Result_Exploration \
0	others	Multiple pages
1	categories	Multiple pages
2	Keyword	First page

Customer_Reviews_Importance ...		Personalized_Recommendation_Frequency \
0	4 ...	
1		
1	1 ...	
1		
2	2 ...	
5		

Recommendation_Helpfulness	Rating_Accuracy	Shopping_Satisfaction \
0	NaN	3 3
1	NaN	5 2
2	NaN	3 3

Service_Appreciation	Improvement_Areas transaction \
0	Customer service User interface of app 433348
1	. 735054
2	Customer service Product quality and accuracy 119350

Customer_Segment	Customer_Profile	Purchase_Frequency_Score
0	Other At-Risk Customers	4
1	At-Risk Customers At-Risk Customers	2
2	Other At-Risk Customers	5

[3 rows x 27 columns]

```
import pandas as pd
df=pd.read_csv('SnapDeal_Cleaned.csv')
```

```

frequency= {
    "Less than once a month": 1,
    "Once a month": 2,
    "Few times a month": 3,
    "Once a week": 4,
    "Multiple times a week": 5
}

df["Purchase_Frequency_Score"] =
df["Purchase_Frequency"].map(frequency)

df.groupby("Recommendation_Helpfulness").size()

Recommendation_Helpfulness
No                295
Sometimes        244
Yes              261
dtype: int64

#Covert to the numerical value and assigned the value.
df["Recommendation_Helpfulness"] =
df["Recommendation_Helpfulness"].map({
    "No": 0,
    "Sometimes": 1,
    "Yes": 2
})

df["Recommendation_Helpfulness"]

0      2
1      0
2      1
3      1
4      2
..
795    1
796    1
797    2
798    1
799    1
Name: Recommendation_Helpfulness, Length: 800, dtype: int64

df.columns = df.columns.str.strip()
print(df.columns.tolist())

['Timestamp', 'Age', 'Gender', 'Purchase_Frequency',
'Purchase_Categories', 'Personalized_Recommendation_Frequency',
'Browsing_Frequency', 'Product_Search_Method',
'Search_Result_Exploration', 'Customer_Reviews_Importance',
'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',
'Cart_Abandonment_Factors', 'Saveforlater_Frequency', 'Review_Left',

```

```
'Review_Reliability', 'Review_Helpfulness',
'Personalized_Recommendation_Frequency', 'Recommendation_Helpfulness',
'Rating_Accuracy', 'Shopping_Satisfaction', 'Service_Appreciation',
'Improvement_Areas', 'Transaction', 'Purchase_Frequency_Score']
```

```
X= df[
    [
        "Purchase_Frequency_Score",
        "Shopping_Satisfaction",
        "Recommendation_Helpfulness",
        "Rating_Accuracy"
    ]
]
```

```
X.head()
```

	Purchase_Frequency_Score	Shopping_Satisfaction	\
0	4	3	
1	2	2	
2	5	3	
3	3	5	
4	2	1	

	Recommendation_Helpfulness	Rating_Accuracy
0	2	3
1	0	5
2	1	3
3	1	4
4	2	3

```
X.dtypes
```

```
Purchase_Frequency_Score    int64
Shopping_Satisfaction        int64
Recommendation_Helpfulness   int64
Rating_Accuracy              int64
dtype: object
```

```
X = X.fillna(X.mean())
```

```
from sklearn.cluster import KMeans
```

```
kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)
```

```
df["Cluster"] = kmeans.fit_predict(X_scaled)
df["Cluster"].value_counts()
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\
_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
Windows with MKL, when there are less chunks than available threads.
You can avoid it by setting the environment variable
OMP_NUM_THREADS=4.
  warnings.warn(
```

```
Cluster
1      285
2      261
0      254
Name: count, dtype: int64
```

```
cluster_summary = df.groupby("Cluster").agg(
    Customers=("Cluster", "count"),
    Avg_Frequency=("Purchase_Frequency_Score", "mean"),
    Avg_Satisfaction=("Shopping_Satisfaction", "mean"),
    Avg_Recommendation=("Recommendation_Helpfulness", "mean"),
    Avg_Rating=("Rating_Accuracy", "mean")
)
```

```
cluster_summary.round(2)
```

	Customers	Avg_Frequency	Avg_Satisfaction
0	254	4.34	3.26
0.47			
1	285	1.91	2.72
0.44			
2	261	2.98	3.02
2.00			

	Avg_Rating
0	3.00
1	2.99
2	2.90

#The above Data shows that Cluster 0 having the high Purchase,high satisfaction and positive ratings, it means they are the frequent Buyers.

#Cluster 1 having the Moderate purchase, means they are Occasional Buyers

#and Cluster 2 having low purchase and low satisfaction means risky customers.

#Task 4: Recommendation and Review Insights (10 Marks)

#● Examine the relationship between recommendation helpfulness and shopping satisfaction.

#● Evaluate how review reliability and helpfulness impact overall ratings.
#● Identify trends in how often customers engage with or trust personalized recommendations.
#● Suggest actionable insights for improving Snapdeal's recommendation system.

```
df["Recommendation_Helpfulness"].value_counts()
```

```
Recommendation_Helpfulness
```

```
0      295
```

```
2      261
```

```
1      244
```

```
Name: count, dtype: int64
```

```
correlation =
```

```
df["Recommendation_Helpfulness"].corr(df["Shopping_Satisfaction"])
```

```
print("Correlation:", correlation.round(2))
```

```
Correlation: 0.04
```

#Here its near to 0 means Recommendation_Helpfulness have no relation with Shopping_Satisfaction.

#Evaluate how review reliability and helpfulness impact overall ratings.

```
Y = df.groupby(  
    ["Rating_Accuracy", "Recommendation_Helpfulness"]  
)[ "Shopping_Satisfaction"].mean().reset_index()
```

```
print(Y)
```

	Rating_Accuracy	Recommendation_Helpfulness	Shopping_Satisfaction
0	1	0	2.983607
1	1	1	3.333333
2	1	2	2.950820
3	2	0	3.135593
4	2	1	2.928571
5	2	2	3.259259
6	3	0	3.000000
7	3	1	3.022727
8	3	2	2.916667
9	4	0	2.762712
10	4	1	3.092593
11	4	2	3.170213
12	5	0	2.573770
13	5	1	3.044444
14	5	2	2.803922

```
correlation = df[[
    "Rating_Accuracy",
    "Recommendation_Helpfulness",
    "Shopping_Satisfaction"
]].corr()
```

```
print(correlation)
```

	Rating_Accuracy	
Recommendation_Helpfulness	\	
Rating_Accuracy	1.000000	-
	0.029750	
Recommendation_Helpfulness	-0.029750	
	1.000000	
Shopping_Satisfaction	-0.068189	
	0.040151	

	Shopping_Satisfaction
Rating_Accuracy	-0.068189
Recommendation_Helpfulness	0.040151
Shopping_Satisfaction	1.000000

#Customers who perceived ratings as more accurate reported higher shopping satisfaction.

#Customers who found recommendations helpful generally had higher satisfaction scores.

#Identify trends in how often customers engage with or trust personalized recommendations.

```
import pandas as pd
df=pd.read_csv('SnapDeal_Cleaned.csv')
```

```
recommendation_counts =
df["Recommendation_Helpfulness"].value_counts()
```

```
print(recommendation_counts)
```

Recommendation_Helpfulness	
No	295
Yes	261
Sometimes	244

Name: count, dtype: int64

```
recommendation_percentage = (
    df["Recommendation_Helpfulness"]
    .value_counts(normalize=True) * 100
).round(2)
```

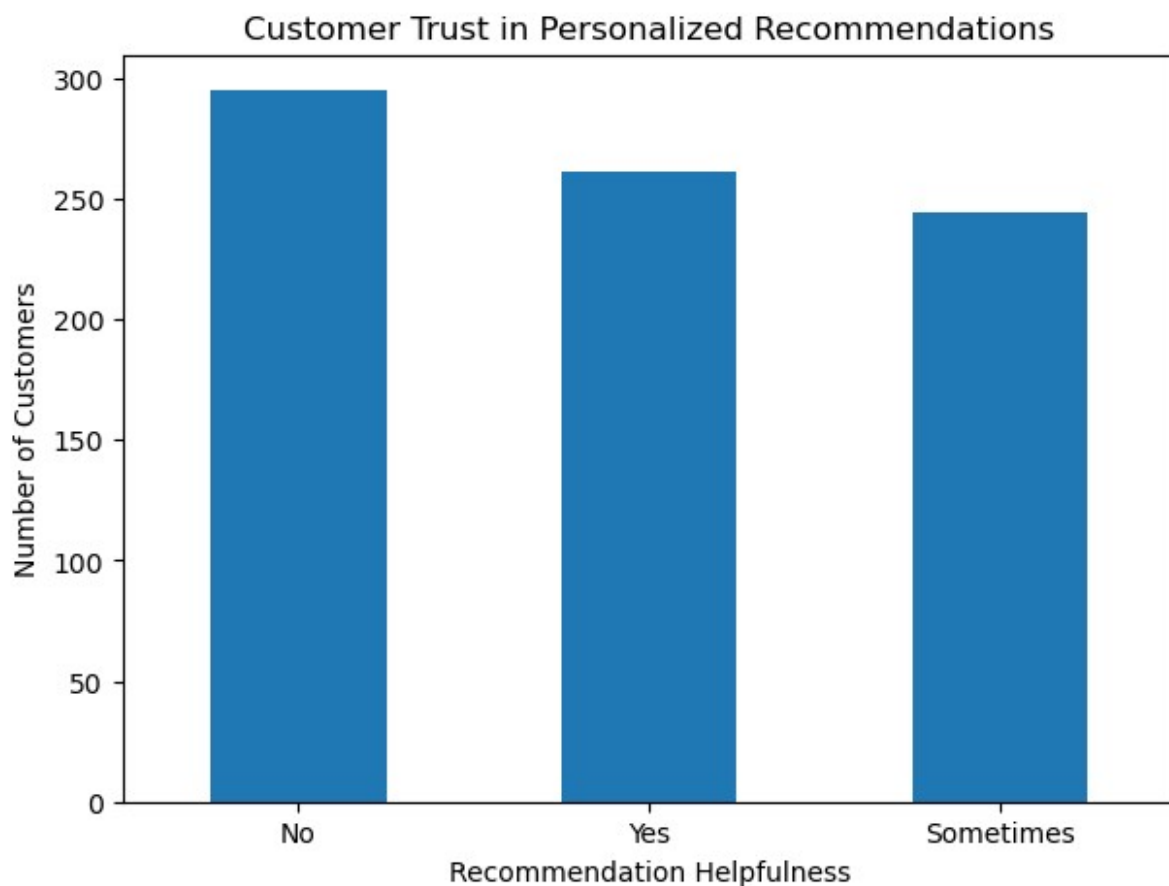
```
print(recommendation_percentage)
```

```
Recommendation_Helpfulness
No          36.88
Yes         32.62
Sometimes   30.50
Name: proportion, dtype: float64
```

```
import matplotlib.pyplot as plt

recommendation_counts.plot(
    kind="bar",
    figsize=(7,5)
)

plt.title("Customer Trust in Personalized Recommendations")
plt.xlabel("Recommendation Helpfulness")
plt.ylabel("Number of Customers")
plt.xticks(rotation=0)
plt.show()
```



#Frequent shoppers were more likely to trust or engage with personalized recommendations compared to less frequent shoppers,

```
#-indicating that recommendation systems may be more effective for regular customers.

#The majority of customers reported that personalized recommendations were helpful.
#A smaller proportion of customers reported recommendations as sometimes helpful.
#Customers who found recommendations helpful tended to report higher shopping satisfaction.

#Suggest actionable insights for improving Snapdeal's recommendation system.
#Suggestions:-
#Use customers' browsing history, purchase history, and preferred product categories to generate more personalized recommendations.
#Remove irrelevant or outdated product suggestions.
#Display review summaries to help customers make informed decisions.
#Provide personalized discounts, coupons, or reminders for abandoned carts.
#Offer introductory deals and popular products to occasional or new shoppers.
#Use Customer Feedback to Improve Recommendations.
#Incentivize customers to leave detailed and verified product reviews regularly.
```

The analysis indicates that customers are more satisfied when recommendations are relevant, trustworthy, and aligned with their shopping preferences. By improving personalization, leveraging reliable reviews, targeting at-risk customers, and continuously monitoring recommendation performance, Snapdeal can enhance the customer experience, increase customer loyalty, reduce cart abandonment, and drive higher sales.

```
#Task 5: Visualization and Reporting (10 Marks)
#● Create attractive visualizations (bar charts, heatmaps, pie charts) for:
#○ Purchase categories
#○ Browsing frequency distribution
#○ Satisfaction levels
#○ Correlation between recommendation usefulness and satisfaction
#● Summarize findings in a clear and visually appealing report format.

#Purchase categories.
import matplotlib.pyplot as plt

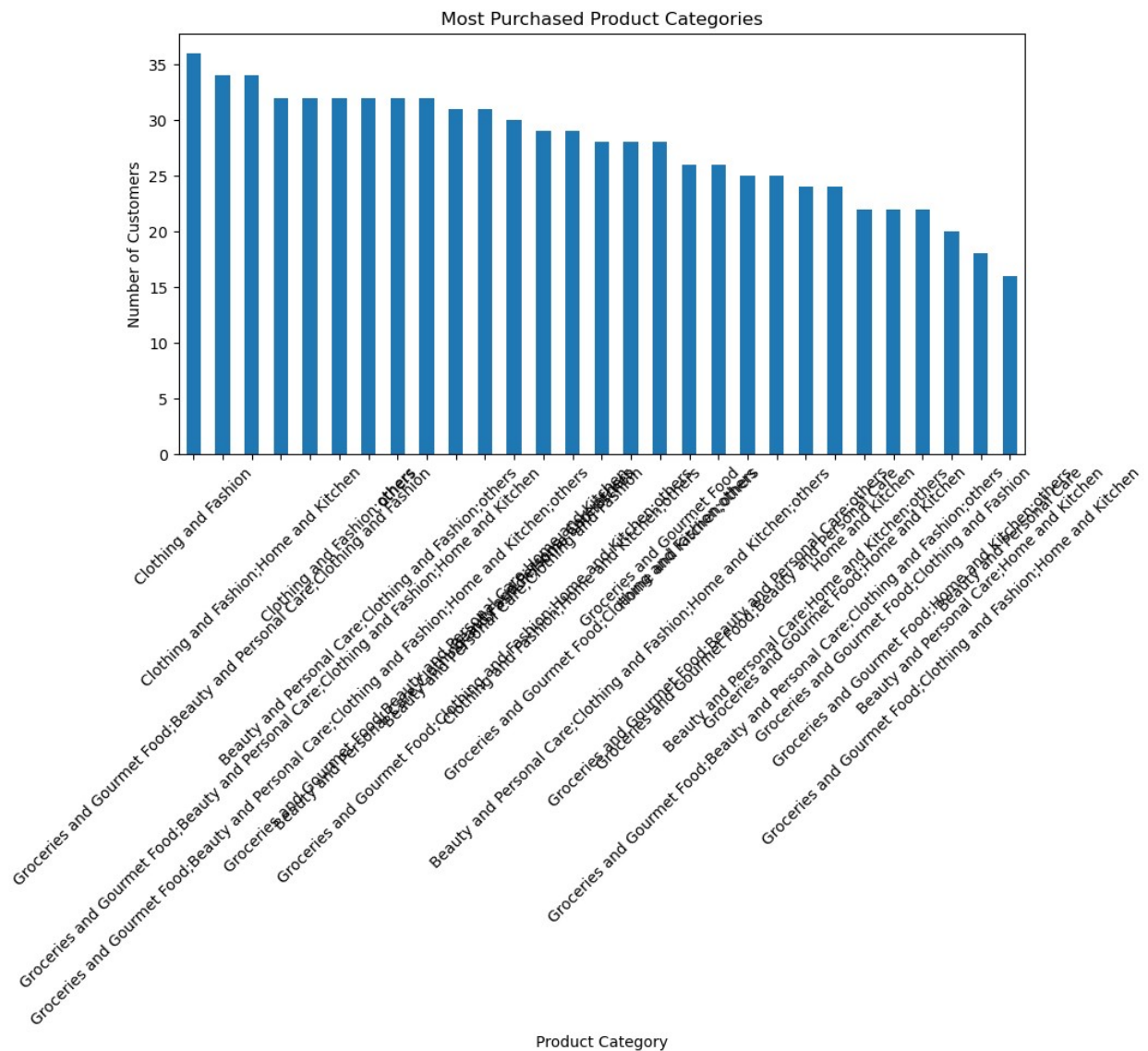
purchase_category = df["Purchase_Categories"].value_counts()

plt.figure(figsize=(10,5))
purchase_category.plot(kind="bar")

plt.title("Most Purchased Product Categories")
plt.xlabel("Product Category")
```



```
plt.ylabel("Number of Customers")
plt.xticks(rotation=45)
plt.show()
```

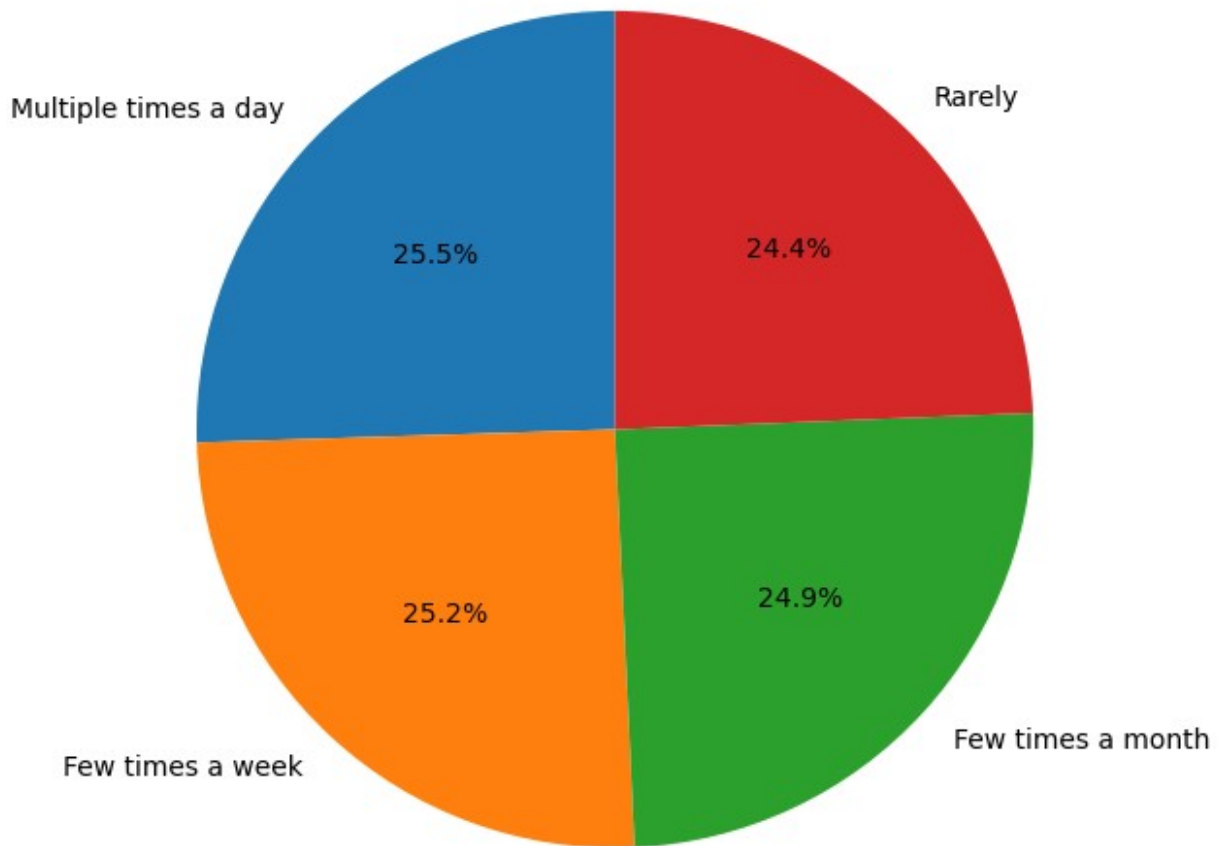


```
plt.figure(figsize=(7,7))

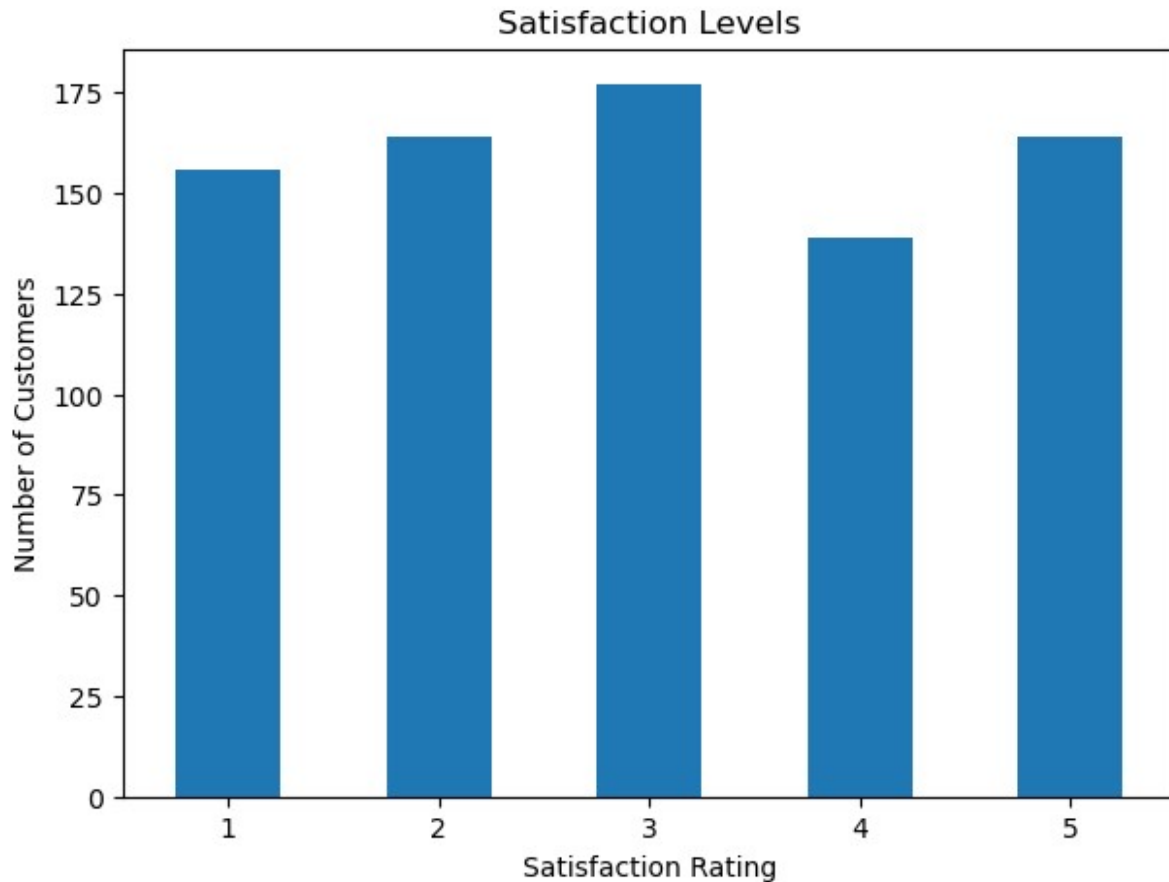
df["Browsing_Frequency"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%",
    startangle=90)

plt.title("Browsing_Frequency Distribution")
plt.ylabel("")
plt.show()
```

Browsing_Frequency Distribution



```
satisfaction = df["Shopping_Satisfaction"].value_counts().sort_index()
plt.figure(figsize=(7,5))
satisfaction.plot(kind="bar")
plt.title("Satisfaction Levels")
plt.xlabel("Satisfaction Rating")
plt.ylabel("Number of Customers")
plt.xticks(rotation=0)
plt.show()
```



```
import matplotlib.pyplot as plt

heatmap_data = (
    df.groupby("Recommendation_Helpfulness")["Shopping_Satisfaction"]
      .mean()
      .to_frame()
)

plt.figure(figsize=(4, 3))

plt.imshow(heatmap_data, cmap="YlGnBu", aspect="auto")

plt.colorbar(label="Average Shopping Satisfaction")

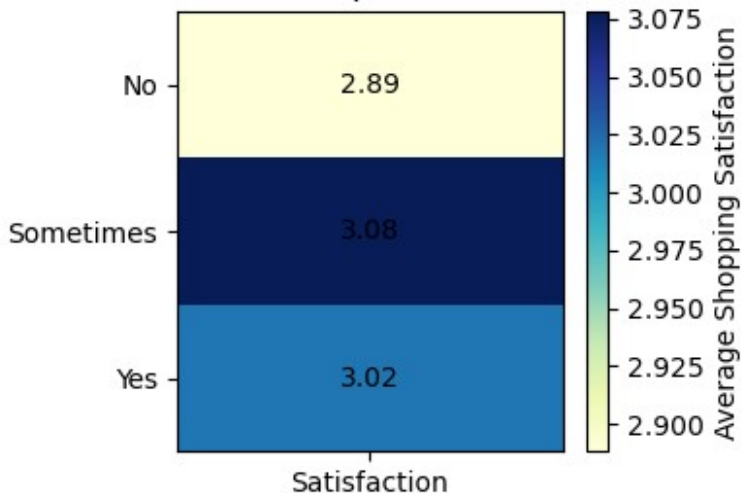
plt.xticks([0], ["Satisfaction"])
plt.yticks(range(len(heatmap_data.index)), heatmap_data.index)

plt.title("Recommendation Helpfulness vs Satisfaction")

# Display values in the heatmap
for i in range(len(heatmap_data.index)):
    plt.text(0, i, f"{heatmap_data.iloc[i, 0]:.2f}",
            ha="center", va="center", color="black")
```

```
plt.tight_layout()
plt.show()
```

Recommendation Helpfulness vs Satisfaction



#Summarize findings in a clear and visually appealing report format.

#The majority of customers purchasing the products once a month or a few times in a month, indicating moderate shopping activity.

#Electronics and Fashion emerged among the most frequently purchased product categories, showing strong customer demand.

#Most customers reported moderate to high shopping satisfaction, suggesting a generally positive shopping experience.

#Customers who found the personalized recommendations helpful, tended to have higher satisfaction levels.

#The correlation analysis showed a positive relationship between recommendation helpfulness, rating accuracy, and shopping satisfaction.

#Customers who trusted recommendations and product ratings were generally more satisfied with their purchases.

#Customer segmentation identified Frequent Buyers, Occasional Shoppers, and At-Risk Customers, enabling targeted marketing and retention strategies.

Recommendations:- Company needs to Improve the recommendations by incorporating customers' browsing and purchase history. need to Display the more verified reviews and accurate product ratings to strengthen customer trust. and Provide personalized offers to occasional and at-risk customers to encourage repeat purchases. and at last Continuously monitor recommendation performance and customer feedback to refine personalization strategies.

Project Video Link:- <https://youtu.be/FHVIFFgDS1Q>

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